Squashed Normal for cont. control

$\pi(a|s,\theta) = \mathcal{N}(a \mid \mu(s,\theta), \sigma^2(s,\theta))$

$a \sim \pi(a|s,\theta) \Rightarrow a \in \mathbb{R}$

$a \in [-1, 1] \Rightarrow \pi(a|s,\theta) \sim ?$

$u \sim \mathcal{N}(u \mid \mu, \sigma^2) \Rightarrow a = \tanh(u)$

$\Leftrightarrow a \sim \text{Sq. Normal}(a \mid \mu, \sigma^2)$

Rep. trick

$$\mathbb{E}_{a \sim \text{Sq. Normal}(a \mid \mu, \sigma^2)} f(a) = \quad \leftarrow \text{differentiable}$$

$$= \mathbb{E}_{z \sim N(0,1)} f(\tanh(\mu + \sigma z)) \to \min_{\mu, \sigma}$$

$$\log \pi(a \mid \mu, \sigma^2)$$

$u \sim p_U(u)$, $a = \tanh(u) = g(u)$ — monotonic one-to-one mapping

$u = g^{-1}(a)$

$$p_A(a) = p_U(g^{-1}(a)) \left| \det \frac{\partial g^{-1}(a)}{\partial a} \right| =$$

$$= p_U(u) \left| \det \frac{\partial g(u)}{\partial u} \right|^{-1}$$

$$\frac{\partial}{\partial u} g(u) = \frac{\partial}{\partial u} \tanh(u) = 1 - \tanh^2(u)$$

$$\log \pi(a \mid \mu, \sigma^2) = \log \mathcal{N}(u \mid \mu, \sigma^2) - \log\left(1 - \tanh^2(u)\right)$$

$\tanh(u) \to \pm 1 \Rightarrow 1 - \tanh^2 \to 0 \Rightarrow \log \to \infty$

$$\log' = \frac{1}{1 - \tanh^2} \cdot \cdots \to \infty$$

$$1-\tanh^2(u)=\operatorname{sech}^2(u)=\left(\frac{2}{\exp(u)+\exp(-u)}\right)^2$$

$$\log\left(1-\tanh^2(u)\right)=2\left(\log 2-\log\left(\exp(u)+\exp(-u)\right)\right)=$$

$$=2\left(\log 2-u-\underbrace{\log\left(1+\exp(-2u)\right)}_{\operatorname{softplus}(-2u)}\right).$$

$$H(\pi) = -\mathbb{E}_{\pi} \log \pi(u \mid \mu, \sigma^2) =$$

$$= -\mathbb{E}_{\mathcal{N}(u \mid \mu, \sigma^2)} \log \mathcal{N}(u \mid \mu, \sigma^2) + \mathbb{E}_{\mathcal{N}(u \mid \mu, \sigma^2)} 2\,(\log 2 -$$

$$- u - \mathrm{softplus}(-2u)) = \frac{1}{2} \log 2\pi + \log \sigma + \frac{1}{2} +$$

$$+ \mathbb{E}_{z \sim \mathcal{N}(0,1)} 2\,(\log 2 - \mu - \sigma z - \mathrm{softplus}(-2(\mu + \sigma z)))$$

KL

$$KL(p(x) \,||\, q(x)) = \int p(x) \log \frac{p(x)}{q(x)} dx$$

$$y = g(x)$$

$$KL(p(y) \,||\, q(y)) = \int p(y) \log \frac{p(y)}{q(y)} dy =$$

$$= \int p(x) \left|\det \frac{\partial g(x)}{\partial x}\right|^{-1} \log \frac{p(x) \left|\det \frac{\partial g(x)}{\partial x}\right|^{-1}}{q(x) \left|\det \frac{\partial g(x)}{\partial x}\right|^{-1}} \left|\det \frac{\partial g(x)}{\partial x}\right| dx$$

KL_{target} in PPO

$B = \text{Parallel Roll Out} = \{(s_t, a_t, GAE_t, g_t, \pi_t, \mu_t, \sigma_t)\}$

for epochs:

$KL_B = \emptyset$

for mini-batch b in B:

$$L_\pi = -\sum_{t \in b} \min\left(\frac{\pi(a_t|s_t,\theta)}{\pi_t} GAE_t, \dots\right)$$

$L_v =$

$L_\theta = L_\pi + c_1 L_v - c_2 Ent(\pi)$

$\theta = \text{opt_step}(\nabla_\theta L)$

$$KL\left(\pi(a_t|s_t,\theta) \,\|\, \pi_t\right) \longrightarrow KL_B$$

if $\text{mean}\, KL = \text{mean}(KL_B) \geq \frac{3}{2} KL_{target}$, then break

Soft Actor Critic (SAC)

$\pi(a|s,\theta)$ – stochastic policy

Max-Entropy RL formulation:

$$J(\theta) = \mathbb{E} \sum_t \left(\gamma^t r(s_t, a_t) - \alpha \log \pi(a_t|s_t,\theta) \right) \to \max_\theta$$

$$V^{soft}(s) = \mathbb{E}_{\pi(a|s)}\left(r(s,a) - \alpha \log \pi(a|s,\theta) + \gamma \mathbb{E}_{p(s'|s,a)} V^{soft}(s') \right)$$

$$\cdot \quad Q^{soft}(s,a) = r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} V^{soft}(s')$$

$$V^{soft}(s) = \mathbb{E}_{\pi(a|s)}\left(Q^{soft}(s,a) - \alpha \log \pi(a|s,\theta) \right)$$

$$Q^{soft}(s,a) = r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} \mathbb{E}_{\pi(a'|s')}\left(Q^{soft}(s',a') - \alpha \log \pi(a'|s') \right)$$

If $Q^{soft}(s,a)$ is known, how to find π?

$$\mathbb{E}_{\pi(a|s,\theta)}\left(Q^{soft}(s,a) - \alpha \log \pi(a|s,\theta)\right) \to \max_{\theta}$$

$$\begin{cases} \sum_a \pi(a|s)\left(Q^{soft}(s,a) - \alpha \log \pi(a|s)\right) \to \max_{\pi} \\ \pi(a|s) \geq 0 \ \forall a, \ \sum_a \pi(a|s) = 1 \end{cases}$$

$$\mathcal{L}(\pi, \mu) = -\sum_a \pi(a|s)\left(Q^{soft}(s,a) - \alpha \log \pi(a|s)\right) + \mu\left(\sum_a \pi(a|s) - 1\right)$$

$$\frac{\partial \mathcal{L}(\pi, \mu)}{\partial \pi(a|s)} = -Q^{soft}(s,a) + \alpha\left(1 + \log \pi(a|s)\right) + \mu = 0$$

$$\Rightarrow \log \pi(a|s) = \frac{1}{\alpha} Q^{soft}(s,a) - 1 - \frac{\mu}{\alpha}$$

$$\Rightarrow \pi(a|s) = \frac{\exp\left(\frac{1}{\alpha} Q^{soft}(s,a)\right)}{Z(s)}$$

Can't compute in practice!

$$L_\pi(\theta) = -\mathbb{E}_{\pi(a|s,\theta)}\left(Q^{soft}(s,a) - \alpha \log \pi(a|s,\theta)\right) =$$

$$= \left\{ \pi(a|s) = Sq.Normal(a|\mu,\sigma^2) \right\} =$$

$$= -\mathbb{E}_{z\sim N(0,1)} Q^{soft}(s, \tanh(\mu(s,\theta) + z\sigma(s,\theta))) +$$

$$+ \alpha H(\pi) \to \min_\theta$$

.

$$\begin{cases} -\mathbb{E}_\pi Q^{soft}(s,a) \to \min_\pi \\ -\mathbb{E}_\pi \log \pi(a|s) \geq H_{target} \end{cases}$$

In practice:

$$H_{target} = dim_actions$$

$$L(\pi, \alpha) = -\mathbb{E}_\pi Q^{soft}(s,a) + \alpha\left(H_{target} + \mathbb{E}_\pi \log \pi\right) =$$

$$= -\mathbb{E}_\pi \left(Q^{soft}(s,a) - \alpha \log \pi(a|s)\right) + \alpha H_{target}$$

$$q(\alpha) = \min_\pi L(\pi, \alpha) = \min_\pi \left(-\mathbb{E}_\pi \left(Q^{soft}(s,a) - \alpha \log \pi\right) + \alpha H_{target}\right) =$$

$$= \left\{ \min_x f(x) = -\max_x (-f(x)) \right\} =$$

$$= -\max_{\pi} \left(\mathbb{E}_{\pi} \left(Q^{soft}(s,a) - \alpha \log \pi \right) \right) + \alpha H_{target}$$

Strong duality: $\min_{\pi : -\mathbb{E}_{\pi} \log \pi \geq H} -\mathbb{E}_{\pi} Q^{soft}(s,a) =$

$$= \max_{\alpha \geq 0} q(\alpha) = \max_{\alpha \geq 0} \left(-\max_{\pi} \mathbb{E}_{\pi} \left(Q^{soft}(s,a) - \alpha \log \pi \right) + \alpha H_{target} \right)$$

$$= -\min_{\alpha \geq 0} \left(\max_{\pi} \mathbb{E}_{\pi} \left(Q^{soft}(s,a) - \alpha \log \pi \right) - \alpha H_{target} \right)$$

$$\min_{\pi : -\mathbb{E}_\pi \log \pi \geq H} \mathbb{E}_\pi Q^{soft}(s,a) = \min_{\alpha \geq 0} \left(\max_{\pi} \mathbb{E}_\pi (Q^{soft}(s,a) - \alpha \log \pi) - \alpha H_{target} \right)$$

Fix α, max w.r.t. π

Fix π, min w.r.t. α $\quad \min_{\alpha \geq 0} -\alpha \left(\mathbb{E}_\pi \log \pi + H_{target} \right)$

If $\mathbb{E}_\pi \log \pi + H_{target} > 0 \Rightarrow$ entropy is too small

$\Rightarrow \alpha$ is increasing

If $\mathbb{E}_\pi \log \pi + H_{target} < 0 \Rightarrow$ entropy is too big $\Rightarrow$

$\Rightarrow \alpha$ is decreasing

SAC scheme

use common backbone with two heads

Init $\pi(a|s,\varphi)$, $Q(s,a|\theta_1)$, $Q(s,a|\theta_2)$, $\overline{\theta}_1, \overline{\theta}_2$, $\alpha = 1$, replay buffer $M = \emptyset$

For episodes:

  Init s_0

  For $t = 0, 1, 2, \ldots$

    $a_t \sim \pi(a_t|s_t,\varphi)$, get r_t, s_{t+1}, $(s_t, a_t, r_t, s_{t+1}) \to M$

    For (s, a, r, s') in mini-batch(M):

      $\tilde{\varepsilon} \sim N(0,1)$, $\tilde{u} = \mu(s'|\varphi) + \tilde{\varepsilon}\sigma(s'|\varphi)$, $\tilde{a} = \tanh(\tilde{u})$

      $y = r + \gamma\left(\min\left(Q(s', \tilde{a}|\overline{\theta}_1), Q(s', \tilde{a}|\overline{\theta}_2)\right) - \alpha \mathrm{Ent}(\pi(a|s',\varphi))\right)$

$$L_{\theta_1} = L_{\theta_1} + \frac{1}{2}\left(y - Q(s,a|\theta_1)\right)^2$$

$$L_{\theta_2} = L_{\theta_2} + \frac{1}{2}\left(y - Q(s,a|\theta_2)\right)^2$$

$$z \sim N(0,I),\ u = \mu(s|\varphi) + z\sigma(s|\varphi)$$

$$L_\varphi = L_\varphi - \min\left(Q(s,\tanh(u)|\theta_1), Q(s,\tanh(u)|\theta_2)\right)$$

$$L_\alpha = L_\alpha - \underbrace{\text{softplus}(\alpha)}\left(\mathbb{E}_\pi \log\pi + H_{target}\right) + \alpha H(\pi)$$

$$\theta_1 = \text{opt_step}(\nabla_{\theta_1} L_{\theta_1}),\quad \theta_2 = \text{opt_step}(\nabla_{\theta_2} L_{\theta_2})$$

$$\varphi = \text{opt_step}(\nabla L_\varphi),\quad \alpha = \text{opt_step}(\nabla L_\alpha),\quad \tilde{\alpha} = clip(\alpha, a, b)$$

$$\bar{\theta}_1 = EMA(\bar{\theta}_1, \theta_1),\ \bar{\theta}_2 = EMA(\bar{\theta}_2, \theta_2)$$

n-step returns in DQN:

$$\left(s_t, a_t, \sum_{i=0}^{n-1} \gamma^i r_{t+i}, s_{t+n}\right) \rightarrow M$$